

TASK ON LOAD BALANCER

1. Configure Classic Load balancer.

Objective

To create and configure a Classic Load Balancer and connect it to an EC2 instance.

Steps

Step 1

Opened AWS Console → EC2 → Load Balancers.

Step 2

Selected Create Load Balancer → Classic Load Balancer.

Step 3

Configured:

Name: devops-classic-lb

Scheme: Internet-facing

Listener: HTTP : 80

Step 4

Configured the health check:

Protocol: HTTP

Port: 80

Path: /

Step 5

Added the running EC2 instance:

Instance ID: i-0f2833b9a9a8643e8

Step 6

Created the Classic Load Balancer.

Step 7

Verified the instance health.

Result: 1 of 1 instance in service ☒

Load Balancer DNS

devops-classic-lb-641439254.us-east-1.elb.amazonaws.com

Result

Classic Load Balancer was successfully created and the EC2 instance is healthy and in service.

Status: Completed ☒

console.aws.amazon.com/ec2/home?region=us-east-1#CreateCLBWizard:

Click to go back, hold to see history

aws [Search] [Alt+S] United States (N. Virginia) CHVK (634176595009) CHVK

EC2 > Load balancers > Create Classic Load Balancer

Summary

Review and confirm your configurations. [Estimate cost](#)

Basic configuration Edit Name: devops-classic-lb Scheme: Internet-facing	Network mapping Edit VPC: vpc-0810ca47bede75314 Availability Zones and subnets: us-east-1b subnet-0a573a8811abff7e7 asg-pub-subnet2	Security groups Edit default sg-0cf8fbd2738f7c2a1	Listeners and routing Edit HTTP:80
Health checks Edit HTTP:80/ - Timeout: 2 seconds - Interval: 5 seconds - Unhealthy threshold: 2 - Healthy threshold: 10	Instances Edit 1 instance added 1 instance in us-east-1b	Attributes Edit - Cross-zone load balancing: On - Connection draining: On - Connection draining timeout: 300 seconds	Tags Edit -

[Cancel](#) [Create load balancer](#)

CloudShell Agent Toolkit for AWS Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LoadBalancer:loadBalancerArn=devops-classic-lb;tab=listeners

aws [Search] [Alt+S] United States (N. Virginia) CHVK (634176595009) CHVK

EC2 > Load balancers > devops-classic-lb

EC2

- Dashboard
- AWS Global View
- Events
- Instances**
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager
- Images

devops-classic-lb [Actions](#)

Details

Load balancer type Classic	Status 0 of 1 instance in service	VPC vpc-0810ca47bede75314	Date created September 18, 2026, 13:37 (UTC+05:30)
Scheme Internet-facing	Hosted zone Z35SXDOTRQ7X7K	Availability Zones subnet-0a573a8811abff7e7 us-east-1b (use1-az2)	

DNS name [Info](#)
[devops-classic-lb-641439254.us-east-1.elb.amazonaws.com](#) (A Record)

Successfully created load balancer: devops-classic-lb
It might take a few minutes for your load balancer to be fully set up and ready to route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LoadBalancer:loadBalancerArn=devops-classic-lb;tab=listeners

aws [Search] [Alt+S] United States (N. Virginia) CHVK (634176595009) CHVK

EC2 > Load balancers > devops-classic-lb

EC2

- Dashboard
- AWS Global View
- Events
- Instances**
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager
- Images

devops-classic-lb [Actions](#)

Details

Load balancer type Classic	Status 1 of 1 instance in service	VPC vpc-0810ca47bede75314	Date created September 18, 2026, 13:37 (UTC+05:30)
Scheme Internet-facing	Hosted zone Z35SXDOTRQ7X7K	Availability Zones subnet-0a573a8811abff7e7 us-east-1b (use1-az2)	

DNS name [Info](#)
[devops-classic-lb-641439254.us-east-1.elb.amazonaws.com](#) (A Record)

This Classic Load Balancer can be migrated to a next generation load balancer. Migration wizard uses your load balancer's current configurations to create a new load balancer. [Learn more](#) [Launch migration wizard](#)

CloudShell Agent Toolkit for AWS Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

2. Configure Application Load balancer.

Objective

To create and configure an Application Load Balancer (ALB) and route traffic to an EC2 instance running Apache web server.

Steps

Step 1: Create Target Group

Opened AWS Console.

Navigated to EC2 → Target Groups.

Created a new target group named alb-apache-tg.

Selected target type as Instance.

Selected protocol HTTP and port 80.

Configured health check protocol as HTTP with path /.

Registered EC2 instance i-0f2833b9a9a8643e8.

Created the target group.

Step 2: Create Application Load Balancer

Navigated to EC2 → Load Balancers.

Selected Create Load Balancer → Application Load Balancer.

Entered the load balancer name as devops-application-lb.

Selected Internet-facing and IPv4.

Selected VPC vpc-0810ca47bede75314.

Selected two Availability Zones:

us-east-1a

us-east-1b

Configured listener as HTTP : 80.

Forwarded traffic to target group alb-apache-tg.

Reviewed the configuration and created the Application Load Balancer.

Step 3: Verify Target Health

Opened Target Groups → alb-apache-tg.

Verified that the Application Load Balancer devops-application-lb was associated.

Verified the registered EC2 instance.

Target health showed 1 Healthy.

Step 4: Test ALB

Opened the ALB DNS name in Google Chrome.

The application page was successfully displayed.

The page showed: HELLO FROM AUTO SCALED EC2 Apache server is Running

Result

The Application Load Balancer was successfully created and configured. The ALB successfully routed HTTP traffic to the EC2 instance through the target group, and the Apache web application was accessible using the ALB DNS name.

Evidence / Screenshots

Target Group configuration – alb-apache-tg

Application Load Balancer configuration – devops-application-lb

Target Group showing 1 Healthy target

ALB DNS name displaying the Apache application in Google Chrome

Status

Completed Successfully

The image displays two screenshots of the AWS Management Console, showing the configuration and status of an Application Load Balancer named 'devops-application-lb'.

Top Screenshot: The console shows the 'devops-application-lb' page with a green notification banner stating 'Successfully created load balancer: devops-application-lb'. The status is 'Provisioning'.

Bottom Screenshot: The console shows the 'devops-application-lb' page with a green notification banner stating 'Successfully created load balancer: devops-application-lb'. The status is 'Active'.

Configuration Details (from both screenshots):

Property	Value
Load balancer type	Application
Scheme	Internet-facing
Status	Provisioning (Top) / Active (Bottom)
Hosted zone	Z35SXDOTRQ7X7K
VPC	vpc-0810ca47bde75314
Availability Zones	subnet-0a573a8811abff7e7 us-east-1b (use1-az2)
Load balancer IP address type	IPv4
Date created	September 18, 2026, 13:55 (UTC+05:30)

The screenshot shows the AWS Management Console for the 'alb-apache-tg' target group. The left sidebar lists navigation options like Dashboard, AWS Global View, Events, and Instances. The main content area displays the target group details, including its ARN, target type (Instance), protocol (HTTP), port (80), and IP address type (IPv4). A summary table indicates 1 total target, with 1 healthy, 0 unhealthy, 0 unused, 0 initial, and 0 draining. Below the table, there is a section for 'Distribution of targets by Availability Zone (AZ)'.

HELLO FROM AUTO SCALED EC2

Apache server is Running

3. Configure Network Load balancer.

Objective

To configure a Network Load Balancer (NLB) for the application and route traffic to an EC2 instance.

Steps Performed

Opened the AWS Management Console.

Navigated to EC2 → Load Balancers → Create Load Balancer.

Selected Network Load Balancer.

Configured the load balancer name as devops-network-lb.

Selected Internet-facing and IPv4.

Selected the required VPC and Availability Zones.

Configured the listener using TCP port 80.

Attempted to create the Network Load Balancer.

Issue Encountered

While creating the Network Load Balancer, AWS displayed the following error:

OperationNotPermittedException: This AWS account currently does not support creating load balancers. For more information, please contact AWS Support.

Due to this AWS account-level restriction, the Network Load Balancer could not be created successfully.

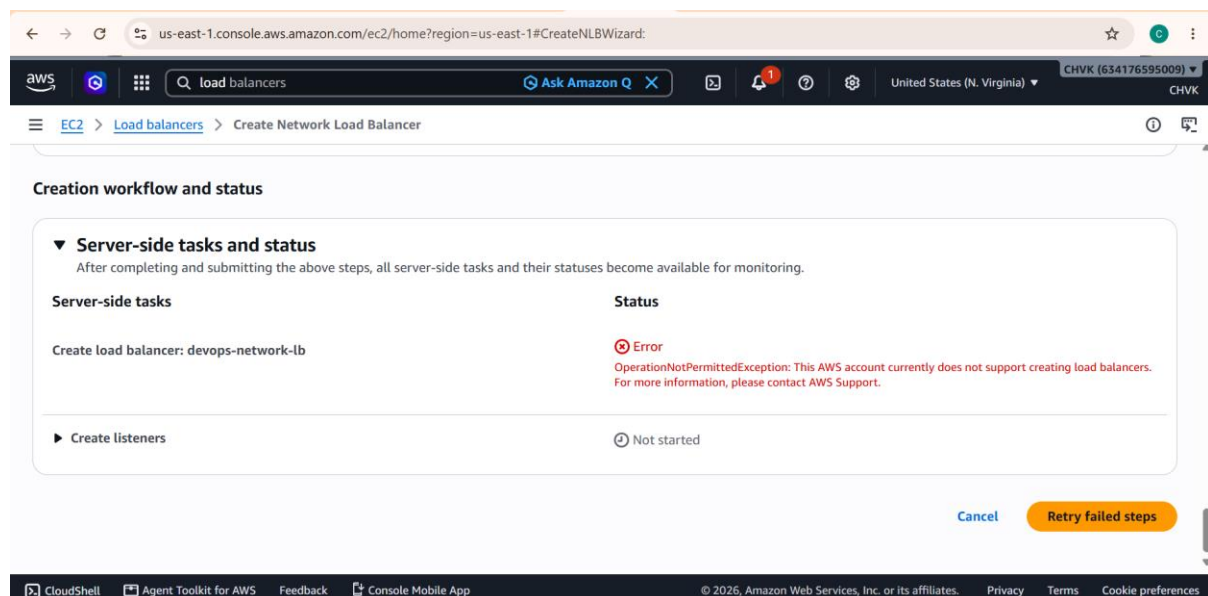
Evidence

Screenshot attached showing the AWS error message during NLB creation.

Result

Task Status: Blocked

The NLB configuration was attempted successfully, but creation could not be completed because the AWS account currently does not support creating load balancers. This issue requires AWS Support/account-level resolution.



4. Attach SSL for application load balancer.

Objective

To attach an SSL certificate to the Application Load Balancer and enable secure HTTPS communication.

Steps Performed

Opened AWS Certificate Manager (ACM) in the us-east-1 region.

Verified the SSL certificate for karthikcloud2026.site.

Confirmed that the certificate status was Issued.

Navigated to EC2 → Load Balancers.

Selected the Application Load Balancer devops-application-lb.

Opened Listeners and rules.

Added a new listener using:

Protocol: HTTPS

Port: 443

Configured the default action to forward traffic to alb-apache-tg.

Selected the ACM certificate for karthikcloud2026.site.

Successfully created the HTTPS listener.

Verification

The Application Load Balancer now has two listeners:

HTTP:80 → alb-apache-tg

HTTPS:443 → alb-apache-tg

Result

SSL was successfully attached to the Application Load Balancer, and HTTPS listener on port 443 was configured successfully.

Evidence / Screenshots

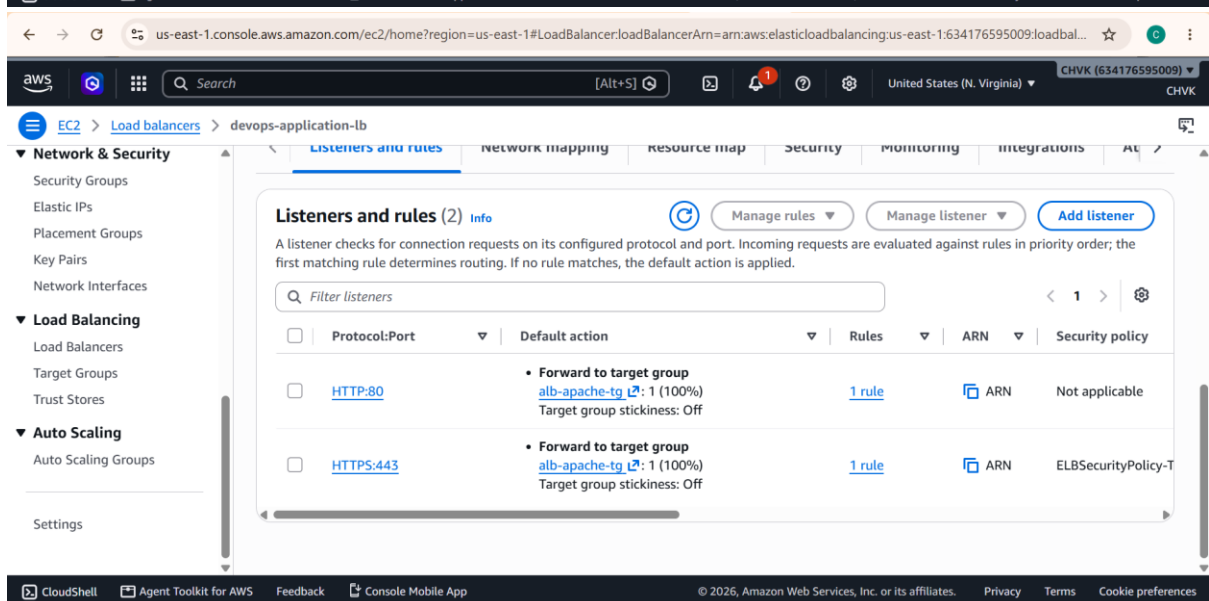
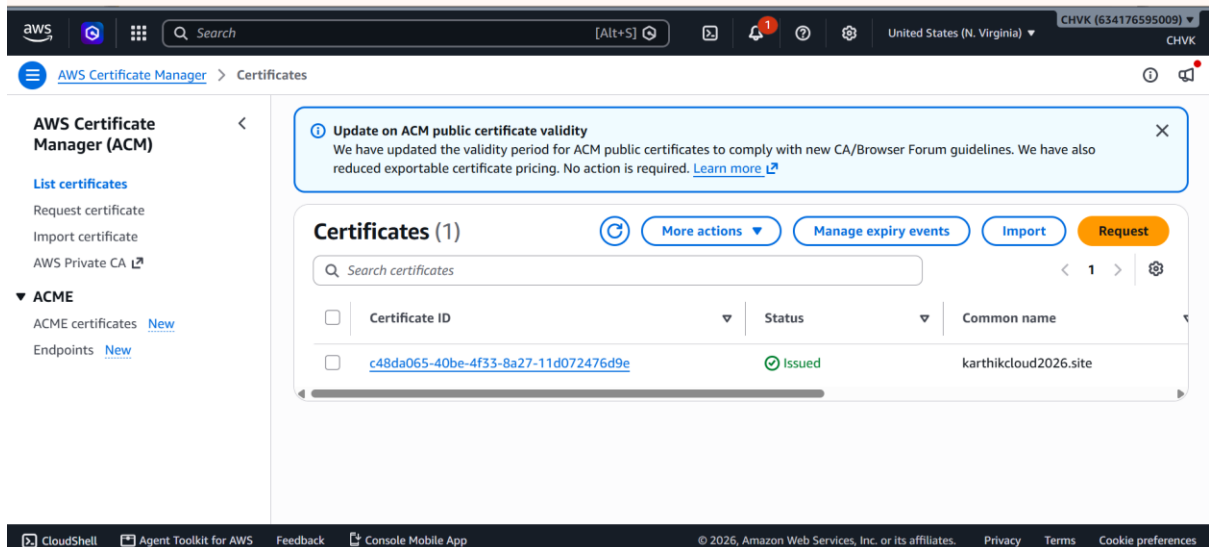
ACM certificate showing Issued status.

ALB showing HTTP:80 and HTTPS:443 listeners.

Target group showing 1 Healthy target.

Status

Completed Successfully



5. Map Application load balancer to R53.

Objective: To map the Application Load Balancer with the Route 53 domain.

Steps Performed:

Opened AWS Console and navigated to Route 53.

Selected Hosted zones and opened karthikcloud2026.site.

Clicked Create record.

Kept Record name blank for the root domain.

Selected Record type as A.

Enabled Alias.

Selected Alias to Application and Classic Load Balancer.

Selected Region as US East (N. Virginia) – us-east-1.

Selected the Application Load Balancer devops-application-lb.

Selected Simple routing.

Clicked Create records.

Verification:

The Route 53 hosted zone displayed the A record with Alias enabled and the target as the Application Load Balancer.

Result:

The domain karthikcloud2026.site was successfully mapped to the Application Load Balancer using a Route 53 Alias A record.

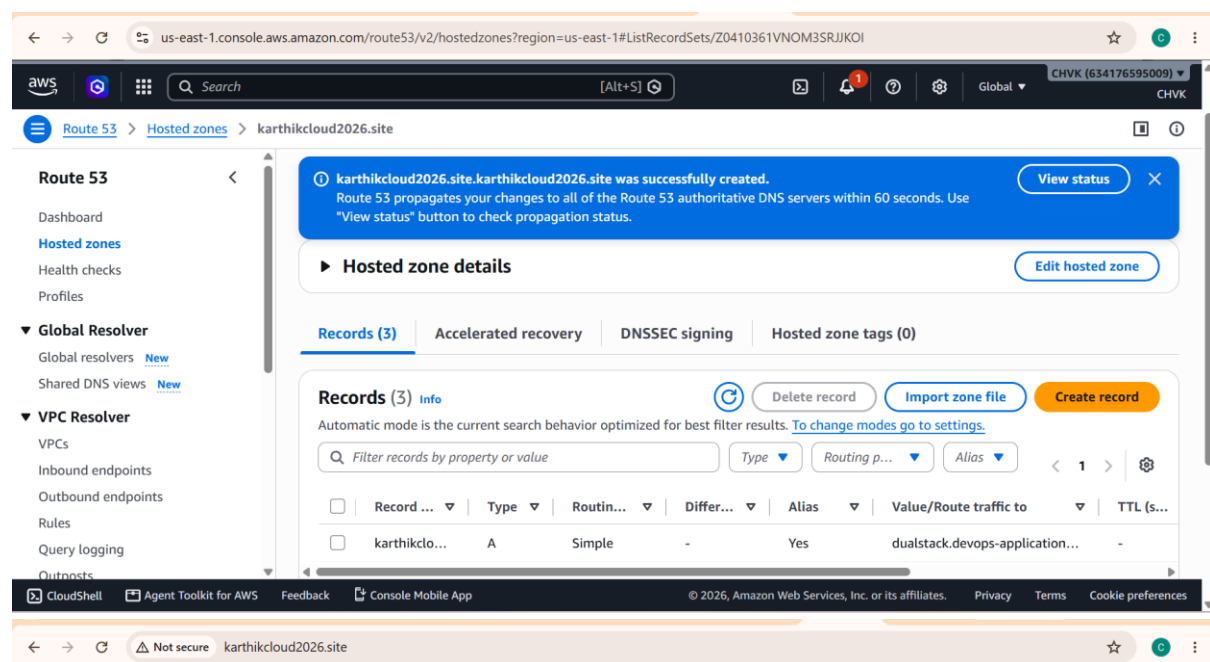
Evidence:

Route 53 Create Record screenshot.

Route 53 successfully created A record screenshot.

Browser domain verification screenshot.

Status: Completed



HELLO FROM AUTO SCALED EC2

Apache server is Running

6. Push the application load balancer logs to S3.

Objective:

To configure Application Load Balancer access logs and store them in an Amazon S3 bucket.

Steps Performed:

Created an S3 bucket named devops-alb-logs-2026-chvk in the US East (N. Virginia) region.

Configured the required S3 bucket policy for ALB log delivery.

Opened EC2 → Load Balancers and selected devops-application-lb.

Navigated to Attributes and tags → Edit load balancer attributes.

Enabled Access Logs.

Provided the S3 URI:

s3://devops-alb-logs-2026-chvk

Clicked Save changes.

Verified the S3 bucket after accessing the Application Load Balancer.

The AWSLogs/ folder appeared in the S3 bucket, confirming that ALB logs were successfully delivered.

Result:

Application Load Balancer access logs were successfully configured and delivered to the S3 bucket.

Evidence:

S3 bucket screenshot.

ALB Access Logs configuration screenshot.

S3 bucket showing AWSLogs/ folder.

Status: Completed

us-east-1.console.aws.amazon.com/s3/buckets/devops-alb-logs-2026-chvk?region=us-east-1&tab=objects

aws

Search

[Alt+S]

1

United States (N. Virginia)

CHVK (634176595009)

CHVK

Amazon S3 > Buckets > devops-alb-logs-2026-chvk

Successfully created bucket "devops-alb-logs-2026-chvk"

devops-alb-logs-2026-chvk

Objects (0)

Copy S3 URICopy URLLownloadOpenDeleteActionsCreate folderUpload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
No objects				
You don't have any objects in this bucket.				

Upload

CloudShellAgent Toolkit for AWSFeedbackConsole Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. PrivacyTermsCookie preferences

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LoadBalancer:loadBalancerArn=arn:aws:elasticloadbalancing:us-east-1:634176595009:loadbal...

aws

load balancers

Ask Amazon Q

1

United States (N. Virginia)

CHVK (634176595009)

CHVK

EC2 > Load balancers > devops-application-lb

Successfully modified load balancer attributes.

Introducing new Capacity Utilization metric for ALB

devops-application-lb

Details

Load balancer type	Status	VPC	Load balancer IP address type
Application	Active	vpc-0810ca47bede75314	IPv4
Scheme	Hosted zone	Availability Zones	Date created
Internet-facing	Z35SXDOTRQ7X7K	subnet-0a573a8811abff7e7 us-east-1b (use 1-az2)	September 18, 2026, 13:55 (UTC+05:30)

CloudShellAgent Toolkit for AWSFeedbackConsole Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. PrivacyTermsCookie preferences

HELLO FROM AUTO SCALED EC2

Apache server is Running

